

Jay Niketan Pathare

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EDUCATION

Masters in Computer Science

University at Buffalo, The State University of New York, Buffalo, NY

August 2025 – December 2026

GPA: 3.85 / 4.0

Bachelor of Engineering, Information Technology

Vivekanand Education Society's Institute of Technology, Mumbai University, India

August 2019 – May 2023

EXPERIENCE

Graduate Student Developer (CSE 611 Project) | Team of 5 | HeinOnline, Buffalo

January 2026 – April 2026

- Engineered a fault-tolerant OCR-LLM pipeline (Tesseract-OCR, Qwen-VL:8b, Qwen-35B) across 3 independent sources – CrossRef API, scrapers, OCR/LLM vision – extracting titles, authors, DOIs, page ranges, ToC metadata, and issue detection from 47 law journals (28,167+ pages).
- Built a fuzzy-matching combinator engine to merge asynchronous JSON outputs into a validated YAML structure, cutting manual data-entry workload by 90% and raising extraction accuracy from 57.03% to 90.81% across five iterations (43 of 47 journals hit 80%+, 8 hit 100%, validated against reference ground truth).
- Designed an "aiscore" system to flag low-confidence fields for human review.

Software Engineer | Thesis Mumbai Tech (Startup), Mumbai, India

August 2024 – August 2025

- Collected requirements directly from stakeholders, translated them into technical specifications, and architected relational databases with linked schemas and dynamic queries for reporting.
- Built and scaled healthcare modules (Patient Management, Consent) supporting 10k+ records, and prototyped a real-time WebSocket + Redis pipeline for IoT baby-warmer data ingestion into PostgreSQL, later replaced by a dedicated IoT device.
- Built a color-blindness screening module leveraging the Ishihara plate test (96–99% established clinical sensitivity for red-green deficiencies), with automated report generation and real-time result visualization for practitioners.
- Integrated secure PDF generation and real-time video conferencing into the platform, and containerized 3+ projects with Docker, cutting setup time 90%.
- Conducted 20+ technical interviews and mentored new hires, contributing to engineering team growth.

Cloud Engineer Intern | Data Maven Pvt Ltd, Mumbai, India

November 2023 – May 2024

- Deployed and configured AWS infrastructure (EC2, RDS) with VPC networking and security groups for secure, low-latency communication.
- Automated resource provisioning, cutting environment setup time 40% and reducing deployment-related issues 25%.

Data Engineer Intern | Go Digital Technology Consulting, Mumbai, India

June 2023 – August 2023

- Cleaned and analyzed datasets using Python (Pandas, NumPy) and MySQL.

TECHNICAL SKILLS

- Languages & Databases:** Python, C, MySQL, PostgreSQL, HTML/CSS
- Frameworks:** Django, Django REST Framework, Django Channels
- AI/ML:** PyTorch, GraphSAGE, GAT, scikit-learn, TensorFlow, CNNs, RNNs, GNNs, Pandas, NumPy
- Software Engineering:** OOP, REST APIs, Full-Stack Development, ETL, Unit Testing, LLM Integration, Microservices, Prompt Engineering, Database Design, Agile, Data Structures & Algorithms, Caching
- Tools:** Docker, Git, GitHub, GitHub Copilot, Tableau, OAuth/Auth0, Locust

ACADEMIC PROJECTS

Decoupled Learning Management System (LMS) | Individual Project

July 2026

- Architected a full-stack LMS (Django REST Framework, React) with Redis-backed real-time messaging via Django Channels for live quizzes and leaderboards; supported 45+ users with sub-500ms latency.
- Implemented idempotent server-side quiz grading, role-based access control for instructors/students, and a backtracking algorithm to generate conflict-free course schedules from section and term constraints.
- Secured the LMS with stateless JWT and custom WebSocket authentication; load-tested with Locust, cutting p95 REST latency 26% and quiz-submission latency from 2000ms to 1500ms.

Fraud Detection using Graph Neural Networks | Team of 3 | University at Buffalo, NY

March 2026

- Built a GraphSAGE-based fraud detection model on the IEEE-CIS dataset, using Focal Loss to handle a severe 27:1 class imbalance and achieving a 0.9259 AUC-ROC (vs. 0.8584 MLP baseline, 0.7928 GAT).
- Converted tabular transaction data into a graph network using shared card, device, and email identifiers, capping node groups to prevent memory explosion while capturing relational fraud rings.

Pintos Operating System – User Programs Layer | Individual Project, University at Buffalo

May 2026

- Engineered the user programs layer for Pintos, a teaching x86 OS kernel in C, designing the process lifecycle: stack setup, execution, parent-child synchronization, and clean termination.
- Developed a system call handler supporting 13 operations for process control and file I/O, with thread-safe file descriptor management for 128 open files per process and executable write-denial protection.
- Solved concurrency via heap-allocated sync, passing 100% of 80 test cases.

CERTIFICATIONS, AWARDS & PUBLICATIONS

- AWS Cloud Practitioner** July 2024
- Certificate of Appreciation:** Event Manager, CSE Workshop “GitHub: Hands-On from Basics to Advanced,” Dept. of Computer Science & Engineering, University at Buffalo. February 2026
- Research Publication:** J. Pathare et al., “Music Genre Classification,” IJRAR, Vol. 10, Issue 2, Apr. 2023. Achieved 97.68% accuracy via audio feature extraction using CatBoost and KNN. [Link](#).